

MP1994V2 Proof Ledger

Mortensen–Pissarides (1994) Lean replication

31 July 2026

Purpose. The milestone tracker records project state. This cumulative ledger records exactly what Lean has proved, the hypotheses used, and the economic meaning of each theorem.

1 M1: Surplus Bellman equation

Paper reference	Section 3, journal page 401, equation (8)
Adequacy status	COMPLETE – GREEN
Lean source	MP1994V2/SteadyState/Surplus.lean

1.1 Mathematical statement

For the actual equilibrium surplus

$$S(\varepsilon) = J(\varepsilon) + W(\varepsilon) - U,$$

Equation (8) is a conditional representation theorem: for every object satisfying **ValueEquilibrium** **P**, Lean proves that the surplus derived from its value functions satisfies, for every idiosyncratic state ε ,

$$(r + \lambda)S(\varepsilon) = p + \sigma\varepsilon - b + \lambda \int \max\{S(x), 0\} dF(x) - \beta\theta q(\theta)S(\varepsilon_u). \quad (8)$$

Here **P.shock** is the probability law of the idiosyncratic shock; its induced CDF is F , and integration with respect to dF is represented in Lean by integration against **P.shock**. Also, $\theta q(\theta)$ is **P.workerMeetingRate** **E.theta**. M1 does not prove existence of a **ValueEquilibrium**.

1.2 Economic interpretation

- $p + \sigma\varepsilon$: current match output.
- b : unemployment opportunity cost.
- $\lambda \int \max\{S(x), 0\} dF(x)$: option value of future idiosyncratic draws, allowing rejection of nonpositive continuation surplus.
- $\beta\theta q(\theta)S(\varepsilon_u)$: the employed worker’s foregone search gain at the upper-support job.

1.3 Exact Lean declarations

Role	Declaration
Continuation decomposition	MP1994V2.continuationTerm_eq_integral_activeSurplus_sub
Firm surplus share	MP1994V2.ValueEquilibrium.firm_share
Algebraic surplus flow	MP1994V2.ValueEquilibrium.surplus_flow_equation
Minimal probability theorem	MP1994V2.ValueEquilibrium.surplus_bellman_of_probability
Paper-level wrapper	MP1994V2.ValueEquilibrium.surplus_bellman

1.4 Assumptions

Used:

- equation (3), through the definition of actual surplus;
- equation (4), only at the upper state;
- equations (5) and (6), added to cancel the wage;
- equation (7), subtracted to account for unemployment value;
- probability normalization of `P.shock`;
- `ValueEquilibrium.active_surplus_integrable`.

The minimal theorem assumes only `[IsProbabilityMeasure P.shock]`. The wrapper takes `D : ShockAssumptions P`, installs `D.isProbability` locally, and uses no other field of `D`.

Not used: parameter signs; `CoreEconomicAssumptions`; upper-support restrictions or maximality; atomlessness; mean zero; unit variance; `ShockNormalizationAssumptions`; matching positivity or monotonicity; `MatchingAssumptions`; equations (1) and (2).

1.5 Readable proof

Write the shared continuation term as

$$C(\varepsilon) = \int [\max\{S(x), 0\} - S(\varepsilon)] dF(x).$$

Adding equations (5) and (6) cancels the wage and sums the Nash weights:

$$r[J(\varepsilon) + W(\varepsilon)] = p + \sigma\varepsilon + \lambda C(\varepsilon).$$

Subtracting equation (7), and using equation (3), gives

$$rS(\varepsilon) = p + \sigma\varepsilon - b + \lambda C(\varepsilon) - \theta q(\theta)[W(\varepsilon_u) - U].$$

Equation (4) at ε_u gives

$$W(\varepsilon_u) - U = \beta S(\varepsilon_u).$$

Probability normalization of `P.shock` and active-surplus integrability give

$$C(\varepsilon) = \int \max\{S(x), 0\} dF(x) - S(\varepsilon).$$

Substitution and moving $-\lambda S(\varepsilon)$ to the left proves (8).

1.6 Adequacy and next dependency

The actual `ValueEquilibrium` surplus is used. Equation (8) is derived, not added as an assumption or field. No cutoff, reservation rule, or affine surplus result is used. It is conditional on an existing `ValueEquilibrium` and does not establish that such an equilibrium exists.

Human-review conclusion. Equation (8) matches the paper and is derived non-circularly from equations (3)–(7). The actual equilibrium surplus is used, with no cutoff, affinity result, or reduced equilibrium. The theorem is explicitly conditional on an existing `ValueEquilibrium`.

M2 uses (8) as the reviewed dependency for the affine-surplus and reservation results recorded next.

Review space:

2 M2: Affine surplus and endogenous reservation cutoff

Paper reference	Section 3; journal p. 401, reservation-property paragraph; journal p. 402, equation (12)
Adequacy status	COMPLETE – GREEN
Lean sources	MP1994V2/SteadyState/AffineSurplus.lean; MP1994V2/SteadyState/Cutoff.lean

2.1 Mathematical statements

For every existing `ValueEquilibrium` P , subtracting equation (8) at two states gives

$$(r + \lambda) [S(\varepsilon_2) - S(\varepsilon_1)] = \sigma(\varepsilon_2 - \varepsilon_1).$$

Under $r > 0$, $\lambda \geq 0$, and $\sigma > 0$,

$$S(\varepsilon_2) - S(\varepsilon_1) = \frac{\sigma}{r + \lambda}(\varepsilon_2 - \varepsilon_1),$$

so S is strictly increasing. Define

$$\varepsilon_d = \varepsilon_u - \frac{(r + \lambda)S(\varepsilon_u)}{\sigma}.$$

Lean proves a unique zero on all of $\mathbb{R}$:

$$S(\varepsilon_d) = 0, \quad S(\varepsilon) = 0 \iff \varepsilon = \varepsilon_d.$$

The reservation rules are

$$\begin{aligned} S(\varepsilon) < 0 &\iff \varepsilon < \varepsilon_d, & 0 \leq S(\varepsilon) &\iff \varepsilon_d \leq \varepsilon, \\ J(\varepsilon) < 0 &\iff \varepsilon < \varepsilon_d, & 0 \leq J(\varepsilon) &\iff \varepsilon_d \leq \varepsilon. \end{aligned}$$

Paper equation (12) and its zero-anchored form are

$$S(\varepsilon) - S(\varepsilon_d) = \frac{\sigma(\varepsilon - \varepsilon_d)}{r + \lambda}, \quad S(\varepsilon) = \frac{\sigma}{r + \lambda}(\varepsilon - \varepsilon_d).$$

The M3 bridge is

$$\max\{S(\varepsilon), 0\} = \frac{\sigma}{r + \lambda} \max\{\varepsilon - \varepsilon_d, 0\}.$$

2.2 Proof strategy and economic separation

The expectation and upper-job search term in equation (8) are common across current states, so they cancel when the two equation instances are subtracted. The resulting positive affine slope constructs the unique mathematical zero.

Economic admissibility is proved separately. Equation (2) and $r > 0$ give $V = 0$; equation (1) then gives

$$q(\theta)J(\varepsilon_u) = c.$$

Positive vacancy cost and meeting imply $J(\varepsilon_u) > 0$. Since $1 - \beta > 0$, the upper-state surplus is positive, and therefore $\varepsilon_d < \varepsilon_u$. Recruiting cost thus places the reservation state strictly below the productivity of a newly formed upper-support job.

2.3 Exact Lean declarations

Role	Declaration
Scaled two-point identity	ValueEquilibrium.surplus_difference_scaled_of_probability
Divided affine identity	ValueEquilibrium.surplus_difference
Strict monotonicity	ValueEquilibrium.surplus_strictMono
Derived cutoff	ValueEquilibrium.reservationCutoff
Cutoff zero	ValueEquilibrium.surplus_reservationCutoff_eq_zero
Unique zero	ValueEquilibrium.existsUnique_surplus_zero
Vacancy value	ValueEquilibrium.vacancy_value_eq_zero
Free-entry product	ValueEquilibrium.vacancy_meeting_mul_upper_firm_value_eq_cost
Upper firm value	ValueEquilibrium.upper_firm_value_pos
Upper surplus	ValueEquilibrium.upper_surplus_pos
Cutoff below upper point	ValueEquilibrium.reservationCutoff_lt_epsUpper
Equation (12)	ValueEquilibrium.equation12
Surplus sign rule	ValueEquilibrium.surplus_neg_iff_lt_reservationCutoff
Firm sign rule	ValueEquilibrium.firm_value_neg_iff_lt_reservationCutoff
Active-surplus bridge	ValueEquilibrium.activeSurplus_eq_slope_mul_positivePart
Capstone	ValueEquilibrium.milestone2_capstone

Capstone zero identification. The capstone explicitly includes

$$\forall \varepsilon, \quad S(\varepsilon) = 0 \iff \varepsilon = \varepsilon_d,$$

rather than merely asserting an unidentified unique zero.

2.4 Assumptions

Core bundle. The affinity and cutoff theorems formally take $A : \text{CoreEconomicAssumptions } P$. Within that bundle their proof terms access `r_pos`, `lambda_nonneg`, and `sigma_pos`. Economic admissibility additionally accesses `c_pos` and `beta_lt_one`, as well as equations (1)–(2) and equilibrium $\theta > 0$.

Matching bundle. Paper-facing admissibility and capstone theorems formally take

$$M : \text{MatchingAssumptions } P.$$

Their proof terms access only `vacancyMeetingRate_pos`. The monotonicity fields remain part of the bundled theorem premise even though they are not referenced by the proof term.

Shock bundle. Wrappers taking $D : \text{ShockAssumptions } P$ access only `D.isProbability` in the M2 affinity and zero arguments. The remaining distributional fields are carried by the paper-level bundle but are not accessed by those proof terms.

Thus the formal paper-facing premises include the complete bundles where shown in the signatures. Within them, the proofs do not access atomlessness, upper-support maximality, shock moment normalization, matching monotonicity, or any elasticity or differentiability restriction.

2.5 Equation (12) dependency

`equation12` follows from the affine-difference theorem evaluated at the derived cutoff. The separate zero theorem is required to simplify (12) to

$$S(\varepsilon) = \frac{\sigma}{r + \lambda}(\varepsilon - \varepsilon_d).$$

2.6 Interpretation, limitations, and next dependency

The positive slope means that higher idiosyncratic productivity raises joint match surplus by the constant factor $\sigma/(r + \lambda)$. The unique zero is the reservation productivity: firms destroy matches below it, are indifferent at it, and retain matches at or above it.

These results are conditional on an existing `ValueEquilibrium`. M2 does not prove existence of that equilibrium or of the joint pair (θ, ε_d) . It proves a unique zero on $\mathbb{R}$ and that the zero lies below ε_u , but not that it belongs to the topological support of the shock law. No lower-support assumption is present; the cutoff may lie below the lower effective support, making destruction probability zero.

Human-review conclusion. The surplus-affinity proof is non-circular, and the cutoff is derived from actual equilibrium values. Mathematical zero existence and economic admissibility remain separate; equations (1)–(2) are used only for admissibility. No M3 result is assumed or proved. The reviewed capstone explicitly identifies every surplus zero with `reservationCutoff`.

M3 may use the active-surplus positive-part formula to derive the continuation-value identity and the job-destruction and job-creation equations. No such integral identity is proved in M2.

3 M3: Continuation value, job destruction, and job creation

Paper reference	Section 3; journal page 401, equations (9)–(10); journal page 402, equation (13)
Adequacy status	COMPLETE – GREEN
Lean sources	<code>SteadyState/Continuation.lean</code> <code>SteadyState/JobDestruction.lean</code> <code>SteadyState/JobCreation.lean</code>
Capstone source	<code>SteadyState/StaticConditions.lean</code>

3.1 Definitions and continuation identity

The foundational expected excess and paper-facing tail option value are

$$H(d) = \int \max\{x - d, 0\} dP.\text{shock}(x), \quad T(d) = \int_d^{\varepsilon_u} [1 - F(x)] dx.$$

The CDF F is induced by `P.shock`; it is not a separate primitive. Lean also defines measure and tail forms of the job-destruction residual, the paper-form job-creation residual, and zero predicates for those residuals.

M2 supplies

$$\max\{S(x), 0\} = \frac{\sigma}{r + \lambda} \max\{x - \varepsilon_d, 0\}.$$

Integrating gives the foundational measure-form continuation identity

$$\int \max\{S(x), 0\} dP.\text{shock}(x) = \frac{\sigma}{r + \lambda} H(\varepsilon_d).$$

Integrability of the positive part is derived from `active_surplus_integrable` and the nonzero positive M2 slope; it is not added as a new assumption.

3.2 Strict-tail layer cake

Mathlib's `MeasureTheory.Integrable.integral_eq_integral_meas_lt` gives

$$H(\varepsilon_d) = \int_{t>0} P.\text{shock}\{x : t < \max(x - \varepsilon_d, 0)\} dt.$$

For $t > 0$, the level set is $(\varepsilon_d + t, \infty)$. Probability normalization proves

$$P.\text{shock}.\text{real}((y, \infty)) = 1 - F(y).$$

The almost-sure upper bound makes the tail vanish beyond ε_u . Truncation and the interval-translation lemma then give

$$H(\varepsilon_d) = T(\varepsilon_d).$$

The strict-tail theorem avoids the paper's informal differentiation and integration-by-parts wording. It also avoids atomlessness: no conversion between weak and strict tails is required.

3.3 Equations (9) and (10)

Substitution in equation (8) produces equation (9):

$$(r + \lambda)S(\varepsilon) = p + \sigma\varepsilon - b + \frac{\lambda\sigma}{r + \lambda}H(\varepsilon_d) - \beta\theta q(\theta)S(\varepsilon_u), \quad (9M)$$

and the paper form replaces $H(\varepsilon_d)$ by $T(\varepsilon_d)$.

Equations (1)–(2) and firm sharing give

$$\beta\theta q(\theta)S(\varepsilon_u) = \frac{\beta c}{1 - \beta}\theta.$$

Evaluating (9M) at the zero-surplus cutoff gives the measure form

$$p + \sigma\varepsilon_d = b + \frac{\beta c}{1 - \beta}\theta - \frac{\lambda\sigma}{r + \lambda}H(\varepsilon_d).$$

Replacing H by T is paper equation (10). The middle term is the worker's foregone search opportunity; the final subtraction is the option value of retaining a temporarily unprofitable job.

3.4 Equation (13)

Independently of equation (10), free entry, firm sharing, and M2 affine surplus at the new-job state give

$$q(\theta)(1 - \beta)\frac{\sigma}{r + \lambda}(\varepsilon_u - \varepsilon_d) = c.$$

Positivity of the denominator factors yields

$$q(\theta) = \frac{c}{1 - \beta} \frac{r + \lambda}{\sigma(\varepsilon_u - \varepsilon_d)}. \quad (13)$$

A smaller productivity gap raises the required vacancy meeting rate. Strict decrease of q implies lower market tightness, while the current weak monotonicity assumption on the worker meeting rate implies a weakly lower worker job-finding rate. M3 does not prove a comparative-static theorem across equilibria.

3.5 Exact Lean declarations

Role	Declaration
Expected excess	<code>Primitives.expectedExcess</code>
Tail probability	<code>Primitives.tailProbability</code>
Tail option value	<code>Primitives.tailOptionValue</code>
Strict-tail/CDF identity	<code>Primitives.measureReal_Ioi_eq_one_sub_cdf</code>
Excess integrability	<code>ValueEquilibrium.positivePart_sub_cutoff_integrable</code>
Layer-cake identity	<code>ValueEquilibrium.expectedExcess_cutoff_eq_tailOptionValue</code>
Active-surplus expectation	<code>ValueEquilibrium.integral_activeSurplus_eq_slope_mul_expectedExcess</code>
Equation (9), measure form	<code>ValueEquilibrium.equation9_measure</code>
Equation (9), paper form	<code>ValueEquilibrium.equation9</code>
Search gain	<code>ValueEquilibrium.search_gain_eq</code>
Equation (10), measure form	<code>ValueEquilibrium.equation10_measure</code>
Equation (10), paper form	<code>ValueEquilibrium.equation10</code>
Destruction residual, measure	<code>Primitives.jobDestructionResidualMeasure</code>
Destruction residual, tail	<code>Primitives.jobDestructionResidual</code>
Destruction witness, measure	<code>ValueEquilibrium.satisfiesJobDestructionMeasure</code>
Destruction witness, tail	<code>ValueEquilibrium.satisfiesJobDestruction</code>
Creation product	<code>ValueEquilibrium.job_creation_product_identity</code>
Equation (13)	<code>ValueEquilibrium.equation13</code>
Creation residual	<code>Primitives.jobCreationResidual</code>
Creation witness	<code>ValueEquilibrium.satisfiesJobCreation</code>
Capstone	<code>ValueEquilibrium.milestone3_capstone</code>

3.6 Assumptions and limitations

Paper-facing theorems formally carry the complete `CoreEconomicAssumptions`, `ShockAssumptions`, and `MatchingAssumptions` bundles.

- Measure-form continuation and equation (9) use probability normalization, active-surplus integrability, and the core fields `r_pos`, `lambda_nonneg`, and `sigma_pos`.
- The tail identity accesses `D.isProbability` and `D.upperSupport`. It does not access `D.noAtoms` or first-moment integrability.
- Its matching bundle is used through `reservationCutoff_lt_epsUpper`, whose proof accesses only `vacancyMeetingRate_pos`. Matching monotonicity remains in the formal bundle but is not referenced by the M3 proof term.
- Equation (10) additionally uses free entry and `beta_lt_one`. Its measure form requires no matching bundle.
- Equation (13) uses free entry, firm sharing, denominator positivity, and cutoff admissibility. Its shock bundle installs only probability normalization for the M2 affine formula.

Shock mean zero, unit variance, `ShockNormalizationAssumptions`, atomlessness, matching elasticity, differentiability, matching monotonicity, and comparative-static assumptions are deliberately unused.

M3 remains conditional on an existing `ValueEquilibrium`. It introduces no reduced-equilibrium structure, reverse reconstruction, joint-equilibrium existence or uniqueness, unemployment stock, or comparative-static theorem.

Human-review conclusion. The strict-tail layer-cake derivation is valid and does not use atomlessness. Equations (9), (10), and (13) match the paper, and equations (10) and (13) are derived independently. The residual results are forward conditional theorems for an existing `ValueEquilibrium`; no reduced equilibrium, reconstruction, or existence theorem is proved.

3.7 Residual admissibility warning

`SatisfiesJobDestruction` and `SatisfiesJobCreation` alone are not a complete reduced equilibrium. M4 must add $\theta > 0$, $d < \varepsilon_u$, and the relevant primitive, distributional, denominator, and other admissibility assumptions. The quotient-form job-creation residual must not be used without proving its denominator nonzero. The product-form job-creation identity is the robust foundational representation.

3.8 Dependency unlocked

M4 may define a `ReducedEquilibrium` only after supplying the missing admissibility contract. The current tail identity obtains integrability from an existing `ValueEquilibrium`. Reverse reconstruction requires a generic proof that $x \mapsto \max\{x - d, 0\}$ is integrable using the assumption

`ShockAssumptions.firstMomentIntegrable.`

M4 also needs, for an arbitrary admissible cutoff d , the generic identity

`expectedExcess d = tailOptionValue d,`

not only the specialization at `E.reservationCutoff`, before constructing value functions from a reduced pair.

4 M4: Reduced-equilibrium reconstruction and equivalence

Paper reference	Section 3, equations (10), (13), journal page 402
Adequacy status	COMPLETE – GREEN
Lean sources	<code>Equilibrium/Reduced.lean</code> ; <code>SteadyState/ReducedAnalytic.lean</code> ; <code>ForwardBridge.lean</code> ; <code>Reconstruction.lean</code> ; <code>Equivalence.lean</code>

4.1 Objects and robust conditions

`ValueEquilibrium P` is the primitive value system. `ReducedEquilibrium P` stores only $\theta > 0$, $d < \varepsilon_u$, measure-form job destruction, and product-form job creation. A full steady state additionally contains unemployment and is deferred to M6. Measure form integrates directly against `P.shock`; product form avoids division by $\varepsilon_u - d$. The CDF-tail equation (10) and quotient equation (13) are derived.

4.2 Generic analysis and reconstruction

`Primitives.positivePart_sub_integrable` uses probability normalization and first-moment integrability. The generic expected-excess/tail theorem additionally uses the almost-sure upper bound to identify the induced-CDF tail integral for every $d \leq \varepsilon_u$. Atomlessness is unused.

For a reduced pair R , Lean reconstructs

$$\begin{aligned}
 S_R(\varepsilon) &= \frac{\sigma}{r + \lambda}(\varepsilon - d), & V_R &= 0, & J_R &= (1 - \beta)S_R, \\
 U_R &= \frac{b + \beta\theta q(\theta)S_R(\varepsilon_u)}{r}, & W_R &= U_R + \beta S_R, \\
 w_R(\varepsilon) &= \beta[p + \sigma\varepsilon] + (1 - \beta)b + \beta c\theta.
 \end{aligned}$$

The wage is the Nash formula, not an arbitrary residual.

4.3 Verification and public declarations

Product-form job creation proves equation (1), while $V_R = 0$ proves (2). The computed surplus gives (3), worker values give (4), and the definition of U_R gives (7). Measure-form job destruction, expected excess, and search gain first derive candidate equation (8); continuation decomposition and the explicit wage then prove (5) and (6).

Principal declarations include:

- `Primitives.SatisfiesJobCreationProduct; ReducedEquilibrium.`
- `Primitives.positivePart_sub_integrable; Primitives.expectedExcess_eq_tailOptionValue.`
- `ValueEquilibrium.toReducedEquilibrium; ReducedEquilibrium.toValueEquilibrium.`
- `ReducedEquilibrium.toValue_toReduced; ValueEquilibrium.wage_eq_nash_formula.`
- `ValueEquilibrium.reconstruction_components; ValueEquilibrium.reconstruction_economic_roundtrip.`
- `valueEquilibrium_nonempty_iff_reducedEquilibrium_nonempty; m4_representation_capstone.`

4.4 Assumptions and exact scope

The forward bridge carries core, shock, and matching bundles; matching enters through M2 cutoff admissibility. Reverse reconstruction carries only core and shock bundles. The reverse constructor uses probability normalization and first-moment integrability, but not upper support. Upper support is needed for the paper-facing CDF-tail form of equation (10). Thus a full assumption bundle may occur in a signature while a proof term accesses only these named fields. Atomlessness, moment normalization, matching monotonicity, elasticity, and differentiability are unused.

The reduced round trip is exact. The value round trip proves equality of θ , V , U , J , W , wage, and surplus. Exact structure equality is not exported because proof-valued Bellman/admissibility fields are not the stable economic interface. Under the maintained bundles, nonemptiness is logically equivalent. Neither side is proved nonempty because no initial witness is constructed. M4 proves neither existence nor uniqueness.

Human-review conclusion. The robust reduced conditions and explicit Nash wage are economically correct. Equations (1)–(7) are verified rather than assumed; the two directions are non-circular; the reduced round trip is exact; and the value round trip preserves every economic object and surplus. The generic excess/tail analysis is valid, and no existence or uniqueness theorem is proved. M4 is **COMPLETE – GREEN**.

Dependency unlocked. M5 may study existence and uniqueness in the two-variable reduced system, knowing every admissible solution reconstructs the primitive value equilibrium.